

SQL Lab Select and Filter Data Exercises

Overview

This document contains a set of SQL exercises designed to help you practice various SQL commands and techniques. You will be working with a hypothetical `movies` table, which contains the following columns:

- `movie_id` (integer)
- `title` (string)
- `genre` (string)
- `release_year` (integer)

Use these exercises to practice selecting data, filtering data, counting records, and working with distinct values.

Exercises

1. Select all columns from the movies table.

- Write a query to retrieve all columns from the `movies` table.

2. Select only the `title` and `release_year` columns from the movies table.

- Write a query to retrieve the `title` and `release_year` columns from the `movies` table.

3. Select all movies released after the year 2000.

- Write a query to retrieve all movies that were released after the year 2000.

4. Select all movies with the genre 'Action'.

- Write a query to retrieve all movies that have the genre 'Action'.

5. Count the total number of movies in the movies table.

- Write a query to count the total number of movies in the `movies` table.

6. **Count the number of distinct genres in the movies table.**
 - Write a query to count the number of distinct genres in the `movies` table.
7. **Select all movies released between 1990 and 2000.**
 - Write a query to retrieve all movies that were released between 1990 and 2000.
8. **Select all movies that are either in the 'Drama' genre or released before 1990.**
 - Write a query to retrieve all movies that are either in the 'Drama' genre or were released before 1990.
9. **Select all movies whose titles start with 'The'.**
 - Write a query to retrieve all movies whose titles start with 'The'.
10. **Select all movies whose titles do not contain the word 'Love'.**
 - Write a query to retrieve all movies whose titles do not contain the word 'Love'.
11. **Select all movies with genres in ('Drama', 'Fantasy', 'Action').**
 - Write a query to retrieve all movies with genres in ('Drama', 'Fantasy', 'Action').
12. **Select the titles of all 'Fantasy' movies released between 2005 and 2015.**
 - Write a query to retrieve the titles of all 'Fantasy' movies released between 2005 and 2015.
13. **Count the number of movies for each genre.**
 - Write a query to count the number of movies for each genre.
14. **Select the first 10 movies released after the year 2000.**
 - Write a query to retrieve the first 10 movies that were released after the year 2000.
15. **Select the movie titles that contain the word 'Star'.**
 - Write a query to retrieve the titles of movies that contain the word 'Star'.
16. **Select the distinct release years of movies.**

- Write a query to retrieve the distinct release years of movies.
17. **Select the genres that have more than 50 movies.**
- Write a query to retrieve the genres that have more than 50 movies.
18. **Select all movies that are not in the 'Fantasy' genre.**
- Write a query to retrieve all movies that are not in the '**Fantasy**' genre.
19. **Select all movies with a release year that is null.**
- Write a query to retrieve all movies with a release year that is null.
20. **Select all movies released in the 1990s.**
- Write a query to retrieve all movies that were released in the 1990s.
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Submission

Please complete each exercise by writing the corresponding SQL query. Submit your queries in a single document for review. Ensure your queries are formatted correctly and provide the expected results based on the `movies` table structure.

If you have any questions or need further assistance, please feel free to ask during the lab session or contact the instructor.

Happy querying!